

# Tehman Hassan

Final Year Computer Engineering Student  
Department of Computer Engineering, University of Engineering  
and Technology Taxila, Pakistan

2023-UET/CP-011  
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## Profile and Bio

Final-year Computer Engineering student at UET Taxila specializing in backend systems, distributed architecture, and AI-powered application development. Experienced in designing and deploying scalable REST APIs, real-time WebSocket systems, and microservices-based architectures using Python, Django, FastAPI, and AWS — validated through a software engineering internship, freelance client deployments, and multiple hackathon-winning projects.

## Education

### Bachelor of Engineering in Computer Engineering

Aug 2023 - Present

University of Engineering and Technology Taxila, Pakistan

Currently in 6th Semester, having an updated CGPA of 3.38

### Matric (Primary) and High School Intermediate

Apr 2017- Oct 2022

Cadet College Choa Saiden Shah Chakwal, Pakistan

A+ Grade · Computing Club · Football Team · Debating Society · Appointment Holder

## Projects

Leading a team of 4 on an Agentic AI-based FYP — scope and architecture currently being finalized.

### Personalized LMS System – 1st Runner Up in AI Hackathon

Project Link: <https://atomcamp.duckdns.org/>

- Built an adaptive learning platform that personalizes student paths in real time based on performance data
- Implemented dynamic quiz difficulty adjustment and instructor analytics dashboard
- Stack: Python, FastAPI, React, OpenAI API, PostgreSQL

### Built “talentgraph.ai”, talent mapping in Global AI Hackathon

Project Link: <https://skillpath.duckdns.org/>

- Built a global skill-mapping platform that clusters talent by niche, location, and market trends
- Integrated AI-based job description parsing to match organizations with relevant talent pools
- Stack: Python, Django, React, OpenAI API, PostgreSQL

### Full Stack Health Platform for Luther Health LTD

Project Link: <https://luther.health/>

- Developed a production-grade health management SaaS for a UK-based doctor with 15+ assessment funnels
- Reduced AI infrastructure costs by architecting an n8n workflow automation layer on AWS
- Stack: Python, Django, AWS EC2, n8n, OpenAI GPT-4, React, Stripe, PostgreSQL

### Single Cycle Processor Datapath in Verilog HDL

Project Link: <https://github.com/Tehman700/Verilog-HDL>

- Implemented a full single-cycle datapath supporting R, I, Load, Store, and Branch instruction types
- Stack: Verilog HDL, Vivado

### Real-Time Asynchronous Chat App Using WebSockets with End-to-End Encryption

Project Link: <https://github.com/Tehman700/Real-Time-Chat-App-2-Way-Communication-WebSockets>

- Built a full-stack async chat application with E2E encryption, OTP verification, and offline message handling
- Stack: Python, Django Channels, WebSockets, PostgreSQL, React

### Potholes Complaint System (Winning Project) in OOP Course Exhibition

- Built an Android app using Azure Vision AI to verify pothole photo authenticity before filing complaints
- Designed a Spring Boot backend with PostgreSQL for complaint tracking and admin resolution management
- Stack: Java, Spring Boot, Azure Vision AI, PostgreSQL

### AI Agent for Teachers (CLO & Bloom's Taxonomy Report Generator)

Project Link: [https://github.com/Tehman700/Dr.Fawad\\_AI\\_Agent\\_Course](https://github.com/Tehman700/Dr.Fawad_AI_Agent_Course)

- Built a Java-based AI agent that analyzes academic documents and auto-generates CLO-aligned reports
- Implemented tree data structures for document processing and vector ID storage for offline reliability
- *Stack: Java, OpenAI GPT-4, PDF generation*

## Work Experience

### Freelance Backend Software Developer

Aug 2025 – Jan 2026

*Fiverr, Freelance Platform*

- Built and deployed production-grade REST APIs using Python, Django REST Framework, and PostgreSQL, serving clients across multiple time zones
- Implemented real-time features using Django Channels and WebSockets for bi-directional communication in client applications
- Designed and maintained CI/CD pipelines on AWS EC2 with Nginx reverse proxy, reducing manual deployment effort and improving release reliability
- Achieved Level 1 Seller status within 2 months, maintaining 5-star client satisfaction through systematic debugging and error handling across AWS and server environments
- Integrated subscription billing flows and activity logging using Django signals for automated audit trails

### Software Engineering Backend Intern

Jul 2025 – Sep 2025

*Inara Technologies, F11 Markaz, Islamabad, Pakistan*

- Architected a full-stack REST API from scratch using Django REST Framework, implementing JWT authentication, role-based access control, and custom exception handling
- Built a nested commenting system with likes, dislikes, and threaded replies across multiple content modules, supporting full CRUD operations
- Developed real-time two-way communication features using WebSockets and Django Channels for live user interaction
- Configured GitHub Actions CI/CD pipelines integrated with AWS deployment infrastructure, cutting manual release steps significantly

## Skills

- **Languages:** Python, Java, JavaScript, Verilog, SQL
- **Frameworks:** Django REST, React, FastAPI, Node.js, Django Channels
- **Cloud & DevOps:** AWS EC2, S3, Nginx, GitHub Actions, Docker
- **Tools:** n8n, Postman, Git, WebSockets
- **Databases:** PostgreSQL, MongoDB

## Competitions

### 1st Runner Up in AI Hackathon at Aurex'26 at Bahria H11

Our team "Titans" won 2nd position in AI Hackathon where we built a personalized Learning Experience LMS using AI

### Winner OOP Course Semester Project

Won the best Semester Project in my department among 150 students, along with certificates and prize money

### Participation in Hackathon in Devsinc

This was a Speed Programming Competition in a Professional Software Company, tackling Data Structures, Greedy Algorithms, and DP Problems

## Certificates

- API Testing Learning Path Certificate
- Postman API Fundamentals Student Expert
- Machine Learning with Python
- 100 Days of Code: The Complete Python Pro Bootcamp
- SQL Basic Hacker rank
- Transformers in LLMs
- Thinking and Problem Solving (LinkedIn)
- Participation in Space Week Organized by NCGSA